Data Science

Introduction to Statistical Inference and R for Data Science

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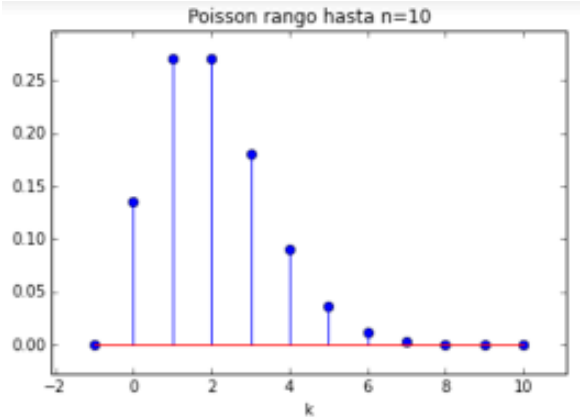
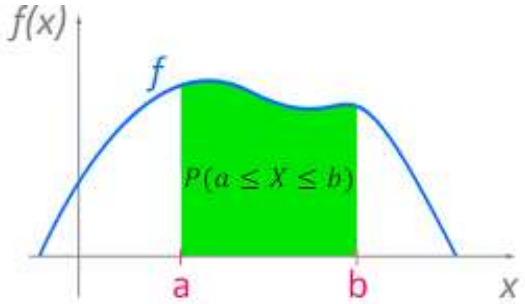
Statistical Inference (intro)

Variable aleatoria: función que asigna un valor numérico a cada resultado de un experimento aleatorio.

V.A. Discretas = Toman valores finitos o contables
V.A. Continuas = Toman valores en un intervalo





$P(a \leq X \leq b) = \int_a^b f(x)dx$ = área bajo la curva de densidad entre a y b

Variable	Descripción	Valores	Tipo
sex	Sexo del estudiante.	"F", "M"	Categórica
age	Edad del estudiante.	15-22 años	Numérica
address	Tipo de residencia.	"U" (urbana), "R" (rural)	Categórica
romantic	Relación romántica.	"yes", "no"	Binaria
freetime	Tiempo libre después de la escuela.	1-5	Numérica (categórica ordenada)

Statistical Inference (intro)

Variable aleatoria: función que asigna un valor numérico a cada resultado de un experimento aleatorio.

X = Los km que hace un coche hasta necesitar una reparación importante

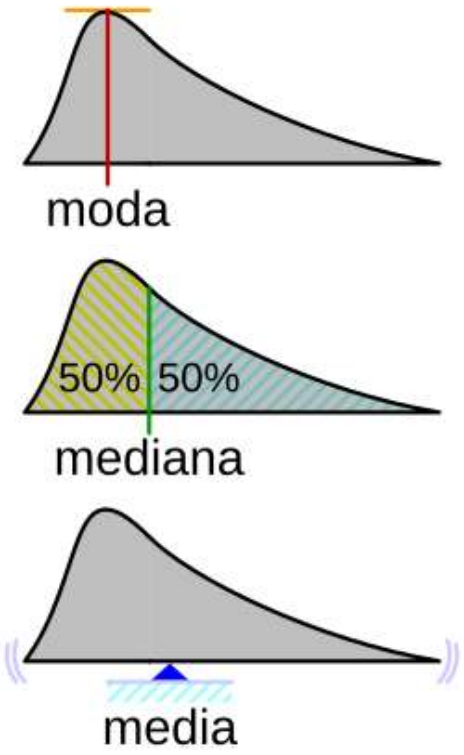
¿Podemos caracterizar X?

- Distribución (si es conocida). En muchos casos por TCL es una Normal.
  - f(x)
  - F(x)
- Características de localización:
  - media
  - mediana
  - …
- Características de dispersión:
  - Varianza
  - Desviación típica
  - …
- …

$$E(x) = \mu = \int_{-\infty}^{\infty} u f(u) du$$

$$mediana(x) = x | F(x) = 0.5$$

$$E((x-\mu)^2) = \sigma^2 = VAR(X) = \int_{\infty}^{\infty} (u-\mu)^2 f(u) du$$



Statistical Inference (Distributions)

- **Bernoulli**

$$p(k) = P(X = k) = \binom{n}{k} p^k (1-p)^{n-k} = \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k}$$

- **Binomial**

$$\mu = E(X) = np$$

$$\sigma^2 = Var(X) = np(1-p)$$

- **Hipergeométrica**

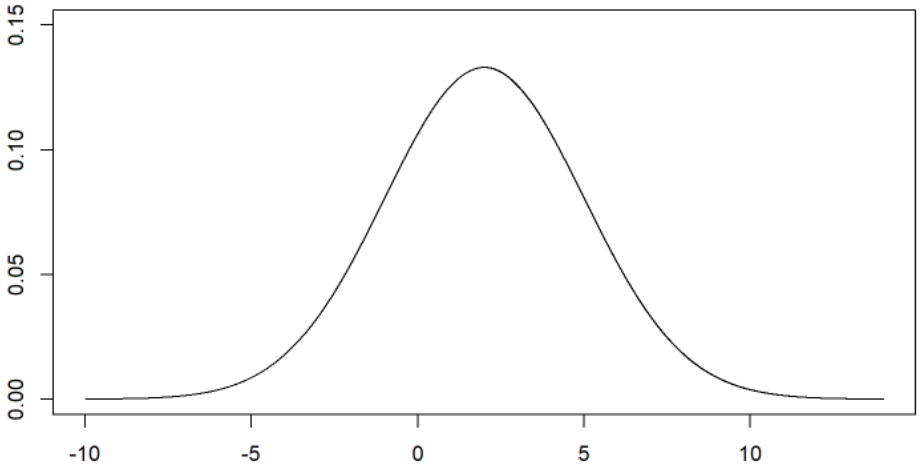
$$f(x) = P(X = x) = \begin{cases} \frac{C_x^R C_{nx}^{N-R}}{C_n^N} & max\{0, n-N+R\} \leq x \leq min\{n, R\} \\ 0 & \text{En caso contrario} \end{cases}$$

$$\mu = E(X) = n\frac{R}{N}$$

$$\sigma^2 = Var(X) = n\frac{R}{N}(1-\frac{R}{N})\frac{N-n}{N-1}$$

- **Normal**

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}(\frac{(x-\mu)}{\sigma})^2}$$



- **Chi-Cuadrado**

$$\chi_n^2 = \sum_n z_i^2$$

- **T-student**

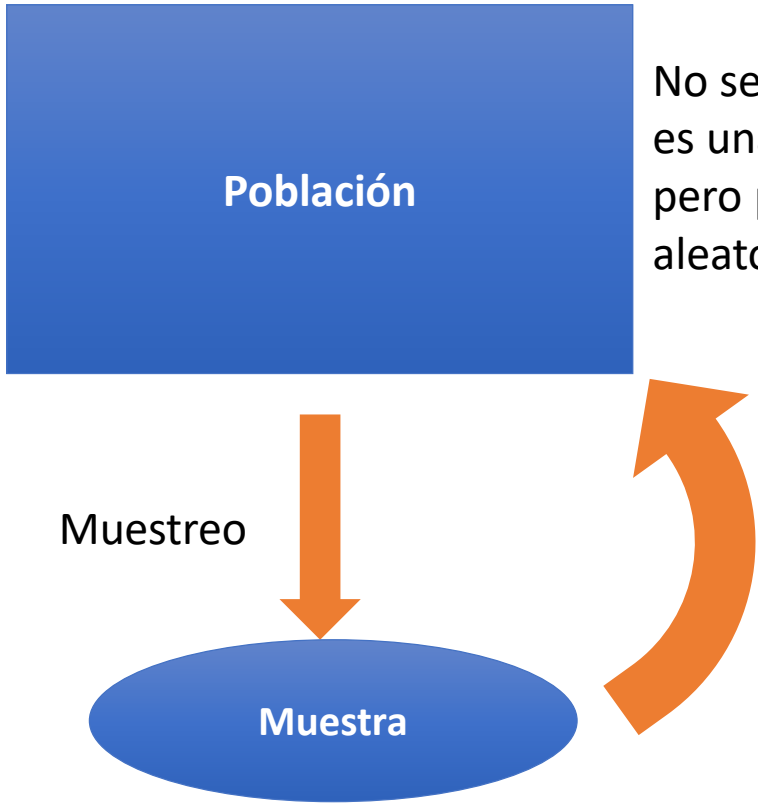
$$t = \frac{Z}{\sqrt{\frac{U}{n}}}$$

- **Distribución F**

$$F_{d_1,d_2} = \frac{U_{1,d_1}/d_1}{U_{2,d_2}/d_2}$$

Statistical Inference

Población de interés. “X = Los km que hace un coche hasta necesitar una reparación importante”



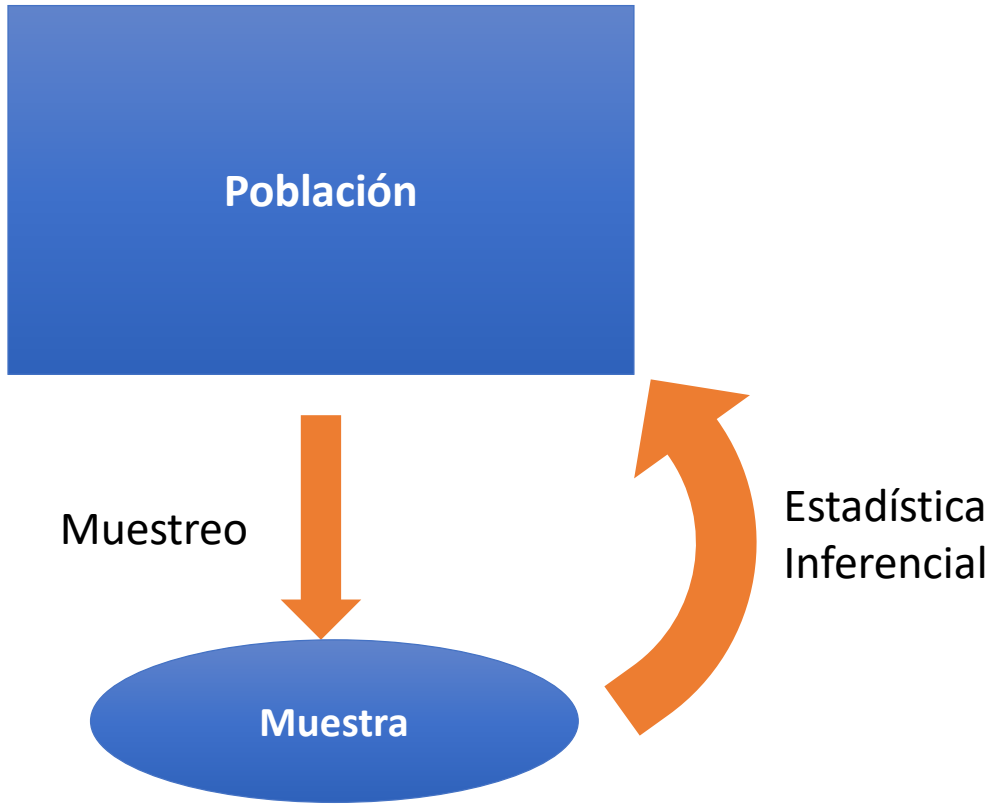
No se que valor va a tener X, porque es una V.A,
pero puedo **caracterizar** esa aleatoreidad

$$N(\mu, \sigma)$$

$$\overline{x} = \sum_{i=1}^{n} \frac{x_i}{n}$$

$$\overline{x} \sim N(\mu_x, \frac{\sigma_x}{\sqrt{n}})$$

Statistical Inference (estimators)



- **Población**: conjunto completo de los elemento que queremos estudiar.
- **Muestra**: subconjunto de la población que observamos y realmente podemos medir.
- **Parámetro**: constante que determina total o parcialmente una distribución y por lo tanto, que caracteriza la población. Generalmente desconocido.
- **Estimador**: función g(x1,x2,…,x3) que depende únicamente de los valores de una muestra y se utiliza para estimar los parámetros desconocidos.

Estimators

- Assume that $x_1, x_2, \ldots, x_n$ are the observed values from n independent and identically distributed (iid) random variables $X_1, X_2, \ldots, X_n$, with density function $f_{Xi}(x,\theta)$.
- We define an estimator of the parameter θ as a function $\hat{\theta}(X_1, X_2, \ldots, X_n)$, which is **also a random variable**.
- Knowing the distribution of each random variable or relying on the central limit theorem, we can determine the distribution of the estimator.
- Empirical value $\hat{\theta}(x_1, x_2, \ldots, x_n)$ is called **estimate** and provides a point estimation for parameters of the population.

$$\overline{x} = \sum_{i=1}^{n} \frac{x_i}{n}$$

$$\overline{x} \sim N(\mu_x, \frac{\sigma_x}{\sqrt{n}})$$

$$s^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \overline{x})^2$$

$$\frac{(n-1)s^2}{\sigma^2} \sim \chi^2_{n-1}$$

Estimators

Método de los momentos: Consiste en igualar los momentos de la distribución de la población con los momentos de la población.

$$E(x) = \mu = \int_{-\infty}^{\infty} u f(u) du$$

$$E((x-\mu)^2) = \sigma^2 = VAR(X) = \int_{\infty}^{\infty} (u-\mu)^2 f(u) du$$

$$\overline{x} = \sum_{i=1}^{n} \frac{x_i}{n}$$

$$s^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \overline{x})^2$$

$$E(\overline{x}) = E(\sum_{i=1}^{n} \frac{x_i}{n}) = \frac{1}{n} E(\sum_{i=1}^{n} x_i) =$$

$$= \frac{1}{n}(E(x_1) + E(x_2) + \ldots + E(x_n)) = \frac{1}{n}(n\mu) = \mu$$

$$VAR(\overline{X}) = VAR(\sum_{i=1}^{n} \frac{x_i}{n}) = (\frac{1}{n})^2 VAR(\sum_{i=1}^{n} x_i) =$$

$$= \frac{1}{n^2}(VAR(X_1) + VAR(X_2) + \ldots + VAR(X_n)) = \frac{1}{n^2}(n\sigma^2) = \frac{\sigma^2}{n}$$

$$E(\hat{\sigma}^2) = E(\frac{1}{n}\sum_{i=1}^{n}(x_i - \overline{x})^2) = \frac{1}{n}E(\sum_{i=1}^{n}(x_i - \overline{x})^2) = \frac{1}{n}E(\sum_{i=1}^{n}(x_i^2 + \overline{x}^2 - 2x_i\overline{x}) =$$

$$\frac{1}{n}E(\sum_{i=1}^{n}(x_i^2) + \sum_{i=1}^{n}(\overline{x}^2) - 2\overline{x}\sum_{i=1}^{n}(x_i)) =$$

$$\frac{1}{n}E(\sum_{i=1}^{n} x_i^2 + n\overline{x}^2 - 2\overline{x}n\overline{x}) = \frac{1}{n}E(\sum_{i=1}^{n} x_i^2 + n\overline{x}^2 - 2n\overline{x}^2) =$$

$$\frac{1}{n}E(\sum_{i=1}^{n} x_i^2 - n\overline{x}^2) = \frac{1}{n}(E(\sum_{i=1}^{n} x_i^2) - E(n\overline{x}^2)) =$$

$$\frac{1}{n}(\sum_{i=1}^{n} E(x_i^2) - nE(\overline{x}^2)) = (*) = \frac{1}{n}(\sum_{i=1}^{n}(\sigma^2 + \mu^2) - n(\mu^2 + \frac{\sigma^2}{n})) =$$

$$\frac{1}{n}(n(\sigma^2 + \mu^2) - n(\mu^2 + \frac{\sigma^2}{n})) = \frac{1}{n}(n\sigma^2 + n\mu^2 - n\mu^2 - n\frac{\sigma^2}{n}) =$$

$$\frac{n-1}{n}\sigma^2$$

Estimators Properties

- **Estimadores centrados o insesgados:** A la diferencia entre el valor esperado del estimador T del parámetro θ y valor verdadero, $E(T) - \theta$, se le llama sesgo del estimador. Si el sesgo es nulo, se dice que es un estimador centrado o sin sesgo para θ.

- **Estimadores eficientes:** Consideremos dos estimadores del mismo parámetro T_1 y T_2. La eficiencia relativa entre ellos se define, independientemente de que sean o no centrados, con el cociente:

$$\frac{E((T_1 - \theta)^2)}{E((T_2 - \theta)^2)} = k$$

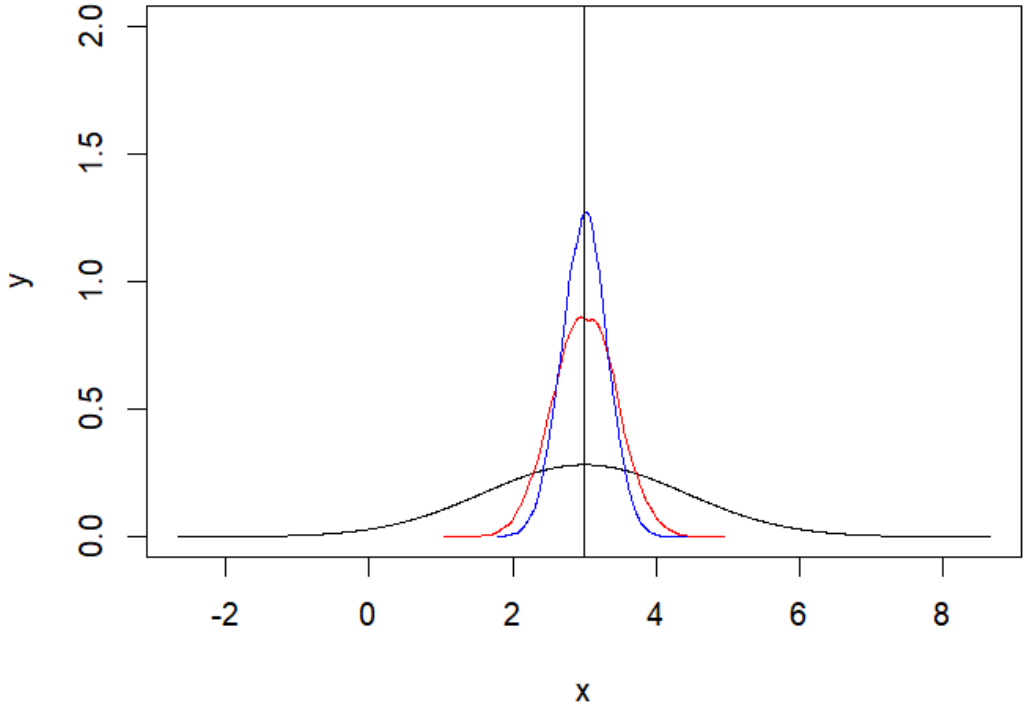
donde $E(T_i - \theta)^2$ es el error cuadrático medio de T_i. Es decir, si T_1 es más eficiente si tiene menos varianza que T_2. Otra manera de verlo: **cuanto menor sea la varianza del estimador, más eficiente es el estimador.**

- **Estimador consistente:** Un estimador es una función de la muestra, por lo tanto, dependerá del tamaño n de la misma. Diremos que un estimador es consistente si al aumentar el tamaño de la muestra, el valor del estimador tiende a converger al valor del parámetro θ. Para ello, el estimador debe ser asintóticamente centrado (cuando el tamaño de la muestra tiende a infinito, la esperanza del estimador tiende al parámetro) y su varianza debe tender a cero.

- **Estimador robusto:** Se dice que un estadístico es robusto si el cambio de un número (pequeño) de mediciones, **sin importar cuán grande sea este cambio**, tiene un efecto limitado en el estadístico.

Estimators Properties

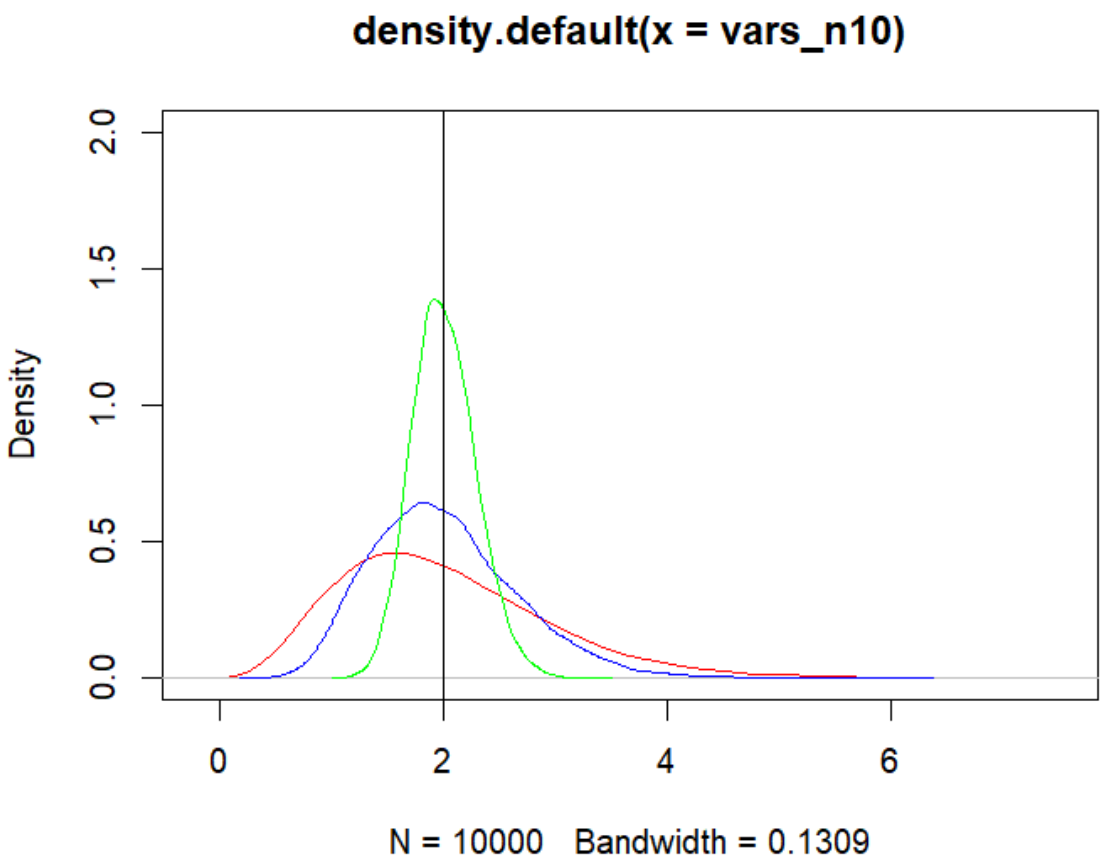
```
# Data
media = 3
varianza = 2

x <- seq(media-4*sqrt(varianza),media+4*sqrt(varianza),0.01)
y <- dnorm(x,mean=media,sd=sqrt(varianza))
plot(x,y,type="l",ylim = c(0,2))

#media es centrado y consistente.

A <- matrix(rnorm(10000*10,mean=media,sd=sqrt(varianza)),nrow=10000)
medias_n10 <- rowMeans(A)
lines(density(medias_n10),col="red")

B <- matrix(rnorm(10000*20,mean=media,sd=sqrt(varianza)),nrow=10000)
medias_n20 <- rowMeans(B)
lines(density(medias_n20),col="blue")
abline(v=media) #se ve que estan centrados
```



El estimador de la media es centrado y consistente

Estimators Properties

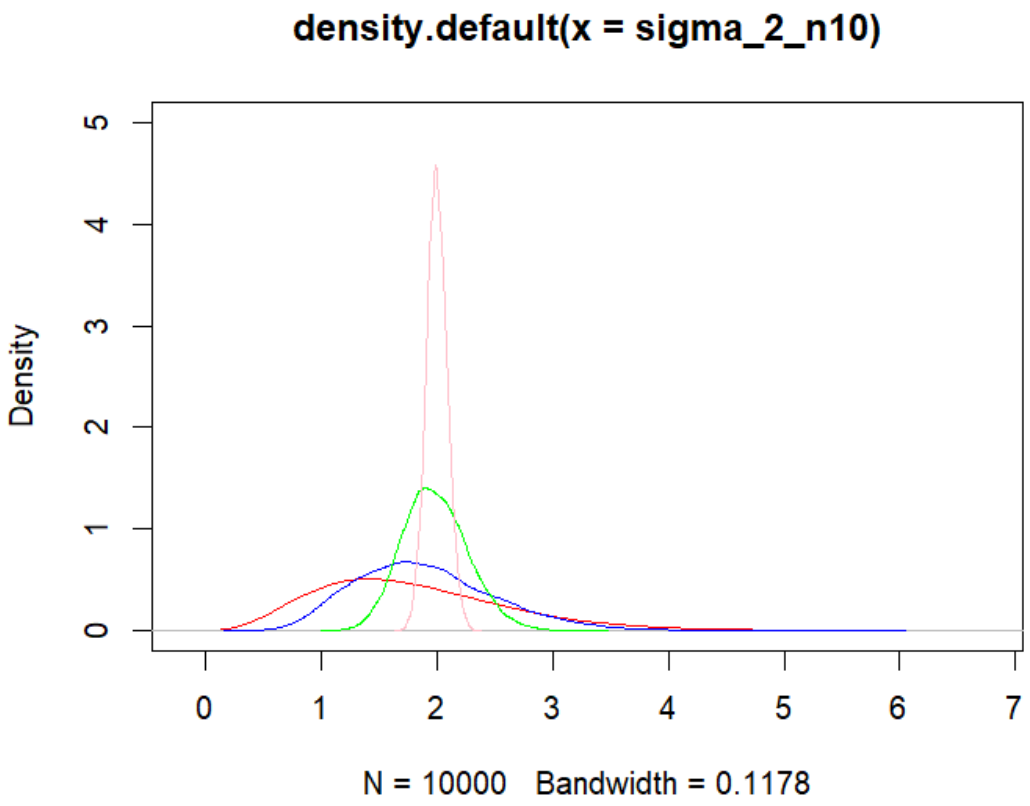
```
#que pasa con la varianza??
library(matrixStats)
vars_n10 <- rowVars(A)
plot(density(vars_n10),col="red",ylim=c(0,2))
mean(vars_n10)

vars_n20 <- rowVars(B)
lines(density(vars_n20),col="blue")
mean(vars_n20)

D <- matrix(rnorm(10000*100,mean=media,sd=sqrt(varianza)),nrow=10000)
vars_n100 <- rowVars(D)
lines(density(vars_n100),col="green")
abline(v=varianza) #se ve que estan centrados
mean(vars_n100)
```



El estimador S^2 de la varianza es centrado y consistente

Estimators Properties

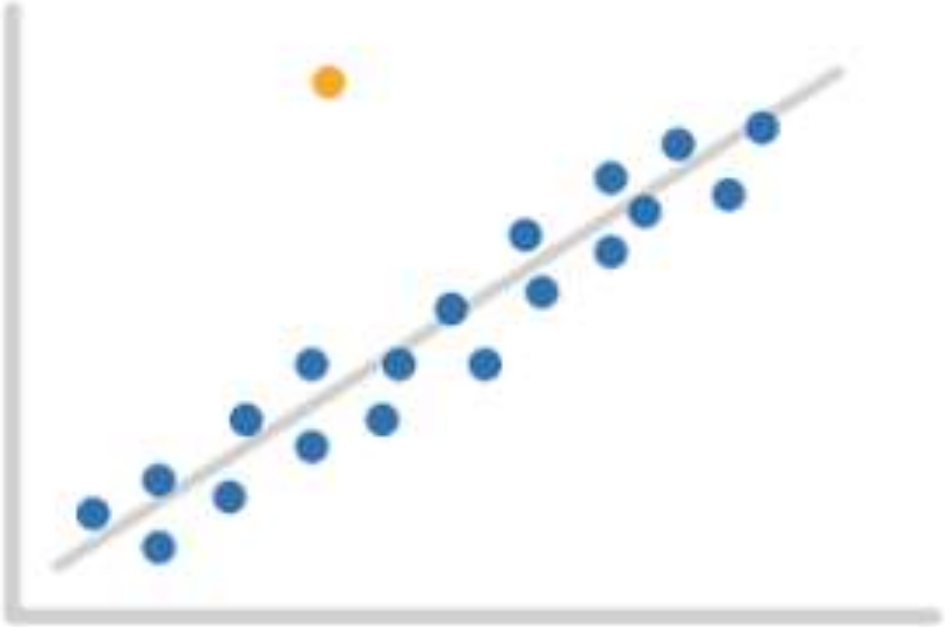
- Crear una función que calcule la varianza con sesgo.
- Crear una matriz E que simule tener experimentos con 1000 observaciones (1000 columnas)
- Utilizar la función apply para aplicar la función a todas las filas de las matrices A, B, D y E.
- Dibujar las funciones de densidad (se observa que al aumentar el tamaño muestral tenemos menos varianza).
- Calcular la media para cada caso y observar que cada vez se parece más al estimador.



El estimador sigma^2 de la varianza es asintóticamente consistente

Estimators Properties

```
> #son robsutos
> random <- rnorm(100,mean=media,sd=sqrt(varianza))
> mean(random)
[1] 3.08067
> var(random)
[1] 2.222337
> #que pasa si metemos un valor atípico (outlier)
> random[1] <- 1000
> mean(random)
[1] 13.05213
> var(random)
[1] 9940.658
```

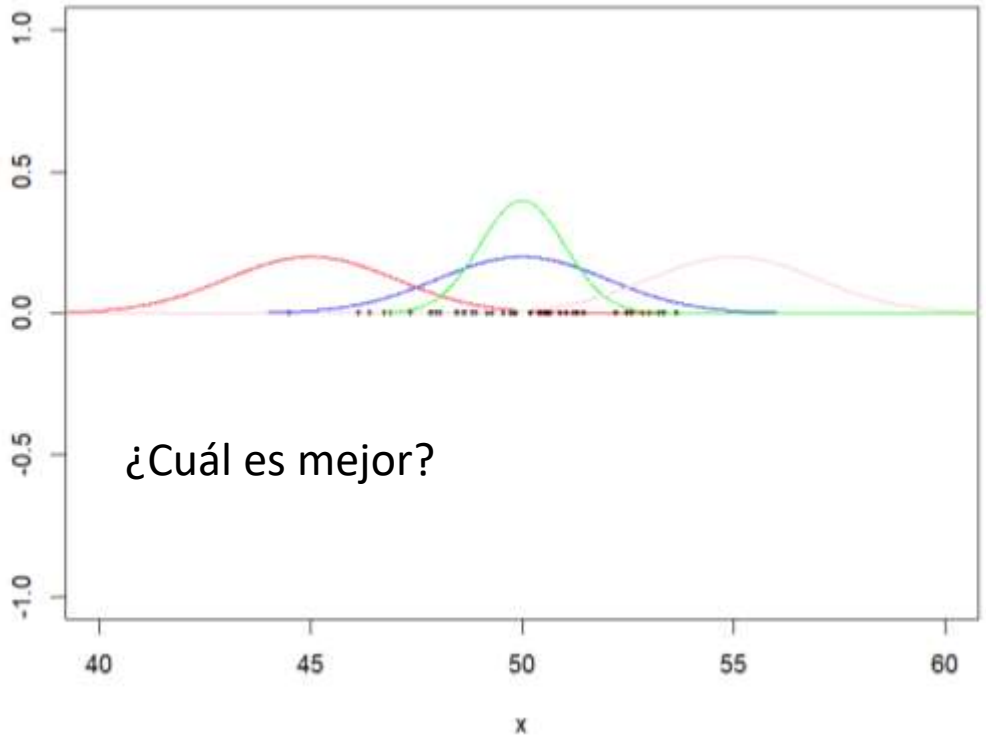


Los estimadores de momentos no son robustos!!

Maximum Likelihood Estimators

$$Max\ L = f(x_1, x_2, \ldots, xn|\theta) = f(x_1|\theta) \cdot f(x_2|\theta) \cdot \ldots \cdot f(x_n|\theta)$$

Buscamos el valor del parámetro que maximice la función de verosimilitud (función de densidad)

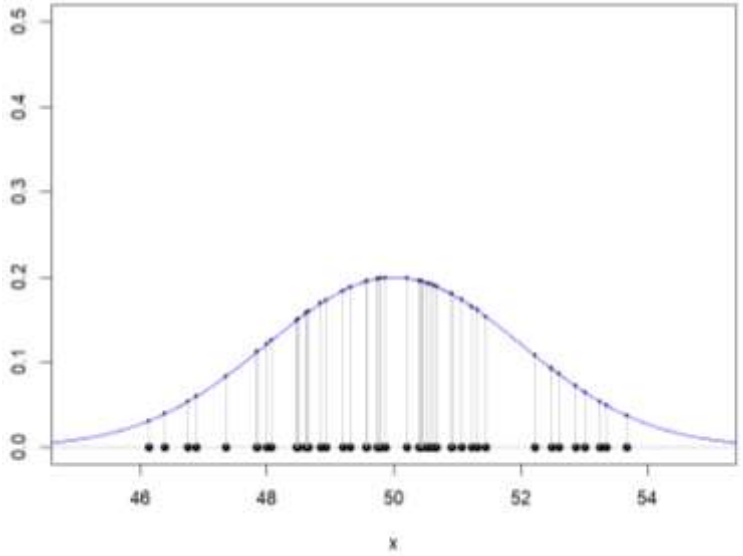


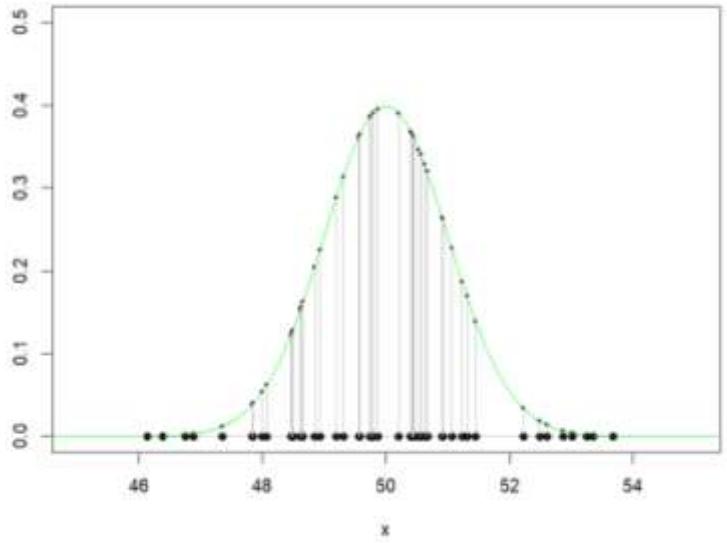
$$f(x;\mu,\sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

$$L(\mu,\sigma^2) = \prod_{i=1}^{n} f(x_i;\mu,\sigma^2)$$

$$L(\mu,\sigma^2) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i-\mu)^2}{2\sigma^2}\right)$$

$$L(\mu,\sigma^2) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2\right)$$





Maximum Likelihood Estimators

$$Max\ L = f(x_1, x_2, \ldots, xn|\theta) = f(x_1|\theta) \cdot f(x_2|\theta) \cdot \ldots \cdot f(x_n|\theta)$$

Buscamos el valor del parámetro que maximice la función de verosimilitud (función de densidad)

$$f(x;\mu,\sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

$$L(\mu,\sigma^2) = \prod_{i=1}^{n} f(x_i;\mu,\sigma^2)$$

$$L(\mu,\sigma^2) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i-\mu)^2}{2\sigma^2}\right)$$

$$L(\mu,\sigma^2) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2\right)$$

$$\ell(\mu,\sigma^2) = \ln L(\mu,\sigma^2) \longrightarrow \ell(\mu,\sigma^2) = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2$$

Estimación de la media

$$\frac{\partial \ell}{\partial \mu} = \frac{\partial}{\partial \mu}\left[-\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln(\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2\right]$$

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i-\mu) = \frac{1}{\sigma^2}\left(\sum_{i=1}^{n} x_i - n\mu\right)$$

$$\frac{1}{\sigma^2}\left(\sum_{i=1}^{n} x_i - n\mu\right) = 0$$

$$\boxed{\hat{\mu}_{MLE} = \frac{1}{n}\sum_{i=1}^{n} x_i = \bar{x}}$$

Maximum Likelihood Estimators

$$Max\ L = f(x_1, x_2, \ldots, xn|\theta) = f(x_1|\theta) \cdot f(x_2|\theta) \cdot \ldots \cdot f(x_n|\theta)$$

Buscamos el valor del parámetro que maximice la función de verosimilitud (función de densidad)

$$f(x;\mu,\sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

$$L(\mu,\sigma^2) = \prod_{i=1}^{n} f(x_i;\mu,\sigma^2)$$

$$L(\mu,\sigma^2) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i-\mu)^2}{2\sigma^2}\right)$$

$$L(\mu,\sigma^2) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2\right)$$

$$\ell(\mu,\sigma^2) = \ln L(\mu,\sigma^2) \longrightarrow \ell(\mu,\sigma^2) = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2$$

Estimación de la Varianza

$$\frac{\partial \ell}{\partial \sigma^2} = \frac{\partial}{\partial \sigma^2}\left[-\frac{n}{2}\ln(\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2\right]$$

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2}\sum_{i=1}^{n}(x_i-\mu)^2$$

$$-\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2}\sum_{i=1}^{n}(x_i-\mu)^2 = 0$$

$$\boxed{\hat{\sigma}^2_{MLE} = \frac{1}{n}\sum_{i=1}^{n}(x_i-\bar{x})^2}$$

Maximum Likelihood Estimators

$$Max\ L = \boldsymbol{f}(x_1, x_2, \ldots, xn|\boldsymbol{\theta}) = \boldsymbol{f}(x_1|\boldsymbol{\theta}) \cdot \boldsymbol{f}(x_2|\boldsymbol{\theta}) \cdot \ldots \cdot \boldsymbol{f}(x_n|\boldsymbol{\theta})$$

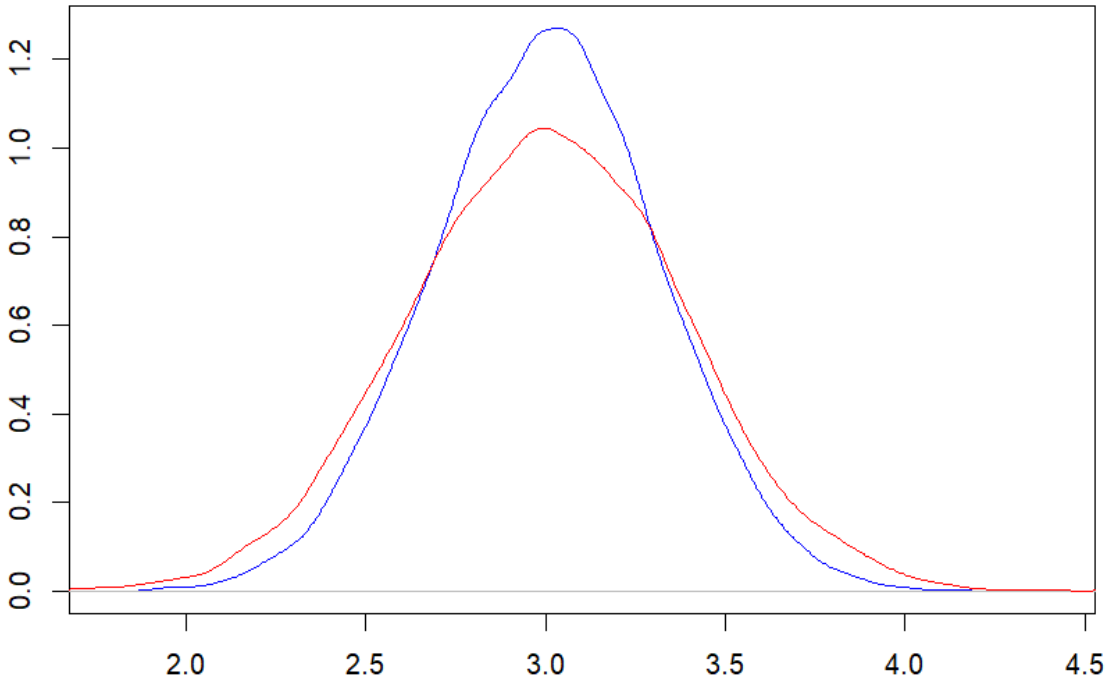
Buscamos el valor del parámetro que maximice la función de verosimilitud (función de densidad)

- No son necesariamente centrados. Ejemplo claro es el EMV de la varianza
- Con frecuencia son consistentes.
- Con frecuencia son eficientes.
- Si existe un estimador centrado de varianza mínima que alcanza la cota de Cramer-Rao, éste es el estimador de máxima verosimilitud y es único.
- La media del estimador tiende al verdadero valor del parámetro y su varianza tiende a la cota de Cramer-Rao.
- Es decir, son los más eficientes.

Estimators

```
> random <- rnorm(100,mean=media,sd=sqrt(varianza))
> mean(random)
[1] 2.935422
> var(random)
[1] 1.745653
> #que pasa si metemos un valor atípico (outlier)
> random[1] <- 1000
> mean(random)
[1] 12.91292
> median(random)
[1] 3.104282
> sd(random)
[1] 99.7145
> mad(random)
[1] 1.360305
> sqrt(varianza) #el valor que queremos estimar
[1] 1.414214
```

```
medianas_n20 <- rowMedians(B)
plot(density(medias_n20),col="blue")
lines(density(medianas_n20),col="red")
```



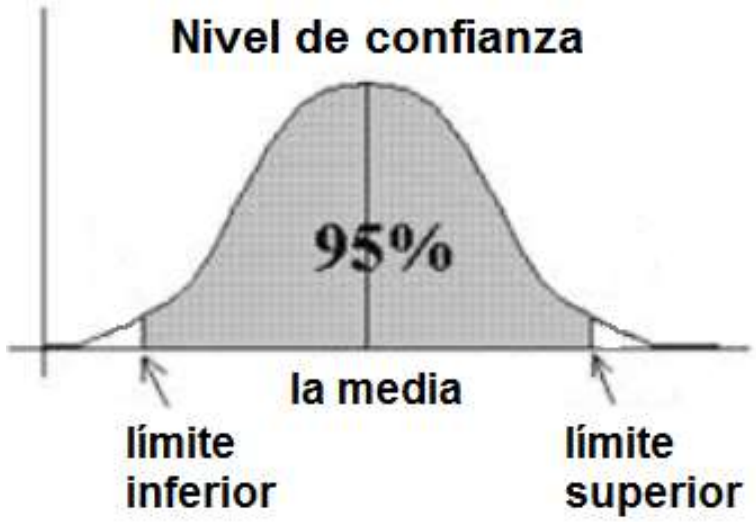
Confidence Intervals

$$\bar{x} \sim N(\mu, \frac{\sigma^2}{n})$$

$$P\{-Z_{\alpha/2} \leq \frac{\bar{x}-\mu}{\sigma/\sqrt{n}} \leq Z_{\alpha/2}\} = 1-\alpha$$

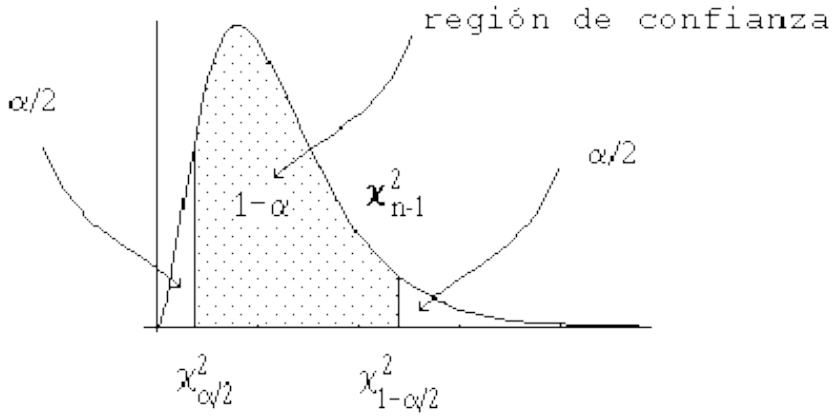
$$\bar{x} - Z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$$

$$\bar{x} - t_{\alpha/2,n-1}s/\sqrt{n} \leq \mu \leq \bar{x} + t_{\alpha/2,n-1}s/\sqrt{n}$$



$$s^2 = \frac{\sigma^2}{n-1}\chi^2_{n-1}$$

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2,n-1}} \leq \sigma^2 \leq \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,n-1}}$$



Introduction

- **R** is a programming language and software environment for statistical computing and graphics

- It is an **open source** project that can be easily extended through user-submitted packages. (9876 packages, as of January 2017)

- Python and R is are the most widely used languages for data mining and data analytics

Introduction

- Being an open source project, there are infinite tutorials available in the internet to learn R.

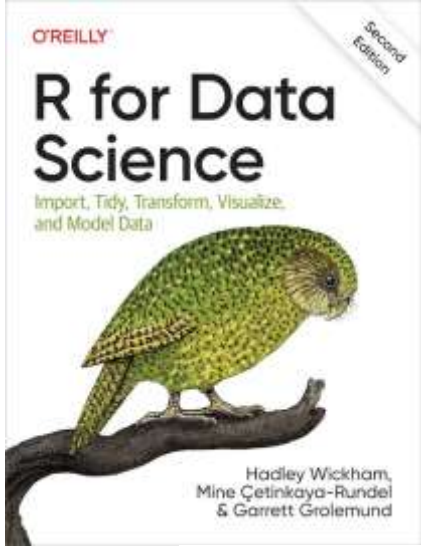
R for Data Science

Garrett Grolemund
Hadley Wickham

https://r4ds.hadley.nz/

Welcome

This is the website for **"R for Data Science"**. This book will teach you how to do data science with R: You'll learn how to get your data into R, get it into the most useful structure, transform it, visualise it and model it. In this book, you will find a practicum of skills for data science. Just as a chemist learns how to clean test tubes and stock a lab, you'll learn how to clean data and draw plots—and many other things besides. These are the skills that allow data science to happen, and here you will find the best practices for doing each of these things with R. You'll learn how to use the grammar of graphics, literate programming, and reproducible research to save time. You'll also learn how to manage cognitive resources to facilitate discoveries when wrangling, visualising, and exploring data.



R packages for data science

The tidyverse is an opinionated **collection of R packages** designed for data science. All packages share an underlying design philosophy, grammar, and data structures.

Install the complete tidyverse with:

```
install.packages("tidyverse")
```

Working with R

- ***R*** can be downloaded from: **https://cloud.r-project.org/**

- We use a Graphic User Interface (GUI) to ease the interpretation of the information we are creating. The GUI is called ***Rstudio***: https://www.rstudio.com/products/rstudio/download/

- ***In ADI tutorial with basic commands***

Working with R

Most of you already know R. In this course, you must write **professional-quality code**.

What this means:

- **Efficient**: Fast execution, smart memory usage
- **Readable**: Clear variable names, logical structure, meaningful comments
- **Reproducible**: Others can understand and run your code
- **Robust**: Handles errors and edge cases appropriately

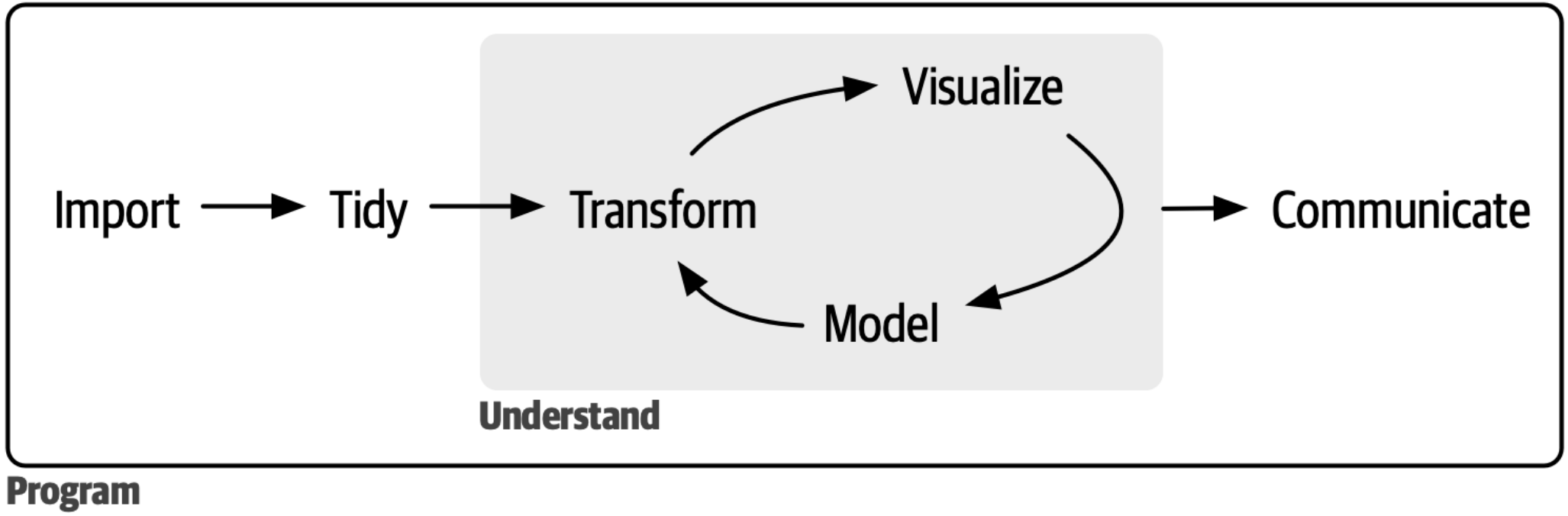
Why it matters: Professional data science requires code that others can review, modify, and trust.
Evaluation: Your assignments will be graded on both correctness AND code quality

Claude

Chat GPT

Gemini

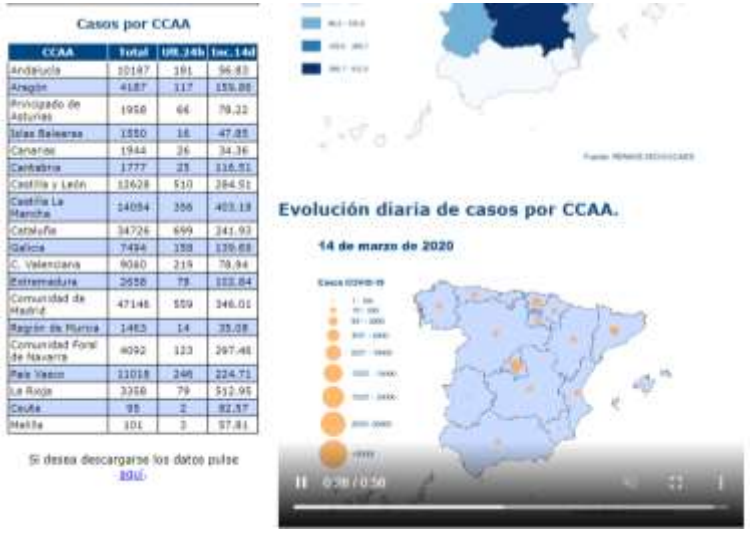
Basic steps



https://r4ds.hadley.nz/intro.html

Basic steps

Import → Tidy /transform → Visualize



- Data1 -> casos_tecnica_provincia.csv
- Data2 -> casos_tecnica_ccaa.csv
- Data3 -> casos_hosp_uci_def_sexo_edad_provres.csv

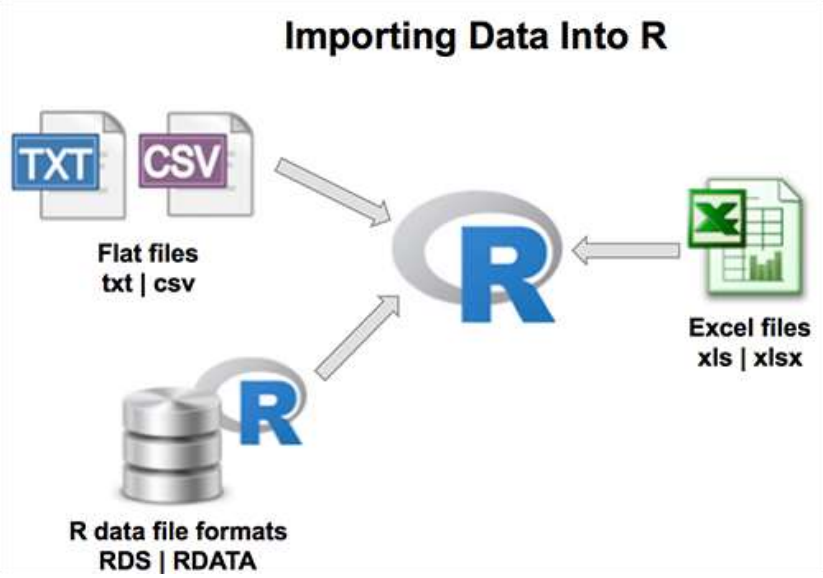
- Data4 -> aeropuertos_2020.json
- Data5 -> airlines_delays.json

- Data6 -> student-mat.csv

Basic steps

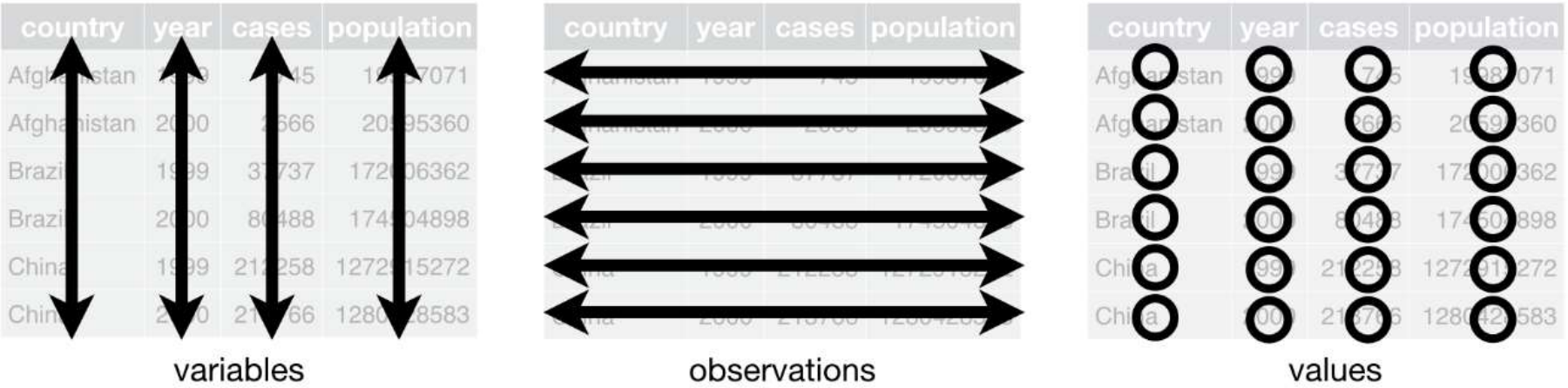
Import → Tidy /transform → Visualize

```
# Número de casos por provincia, fecha y tipo de detección
data1 <- read.csv('data/casos_tecnica_provincia.csv',
            header = T, sep = ',', stringsAsFactors = F,check.names = F)


data3 <- data.table::fread('data/casos_hosp_uci_def_sexo_edad_provres.csv',
            header = T, sep = ',',
            stringsAsFactors = F, check.names = F)


df <- read_csv( simple_csv, col_types = list(x = col_double()) )


myData1 <- fromJSON('./data/aeropuertos_2020.json')
output_dataframe1 <- as.data.frame(myData1$features)
data4 <- output_dataframe1$properties
```



Basic steps

Import → **Tidy /transform** → Visualize



id	bp1	bp2
A	100	120
B	140	115
C	120	125

id	measurement	value
A	bp1	100
A	bp2	120
B	bp1	140
B	bp2	115
C	bp1	120
C	bp2	125

https://r4ds.hadley.nz/data-tidy.html

Basic steps

Import → **Tidy /transform** → Visualize

```
dim(billboard)
head(billboard)

#en este caso, hay dos variables que son:
# semana
# valor
#estos dos valores no están en el formato que queremos

billboard = billboard %>%
  pivot_longer(
    cols = starts_with("wk"),
    names_to = "week",
    values_to = "rank"
  )
head(billboard)
```

artist	track	date.entered	wk1	wk2	wk3	wk4	wk5	wk6	wk7
<chr>	<chr>	<date>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
2 Pac	Baby…	2000-02-26	87	82	72	77	87	94	99
2Ge+h…	The …	2000-09-02	91	87	92	NA	NA	NA	NA
3 Doo…	Kryp…	2000-04-08	81	70	68	67	66	57	54
3 Doo…	Loser	2000-10-21	76	76	72	69	67	65	55
504 B…	Wobb…	2000-04-15	57	34	25	17	17	31	36
98^0	Give…	2000-08-19	51	39	34	26	26	19	2

artist	track	date.entered	week	rank
<chr>	<chr>	<date>	<chr>	<dbl>
2 Pac	Baby Don't Cry (Keep...	2000-02-26	wk1	87
2 Pac	Baby Don't Cry (Keep...	2000-02-26	wk2	82
2 Pac	Baby Don't Cry (Keep...	2000-02-26	wk3	72
2 Pac	Baby Don't Cry (Keep...	2000-02-26	wk4	77
2 Pac	Baby Don't Cry (Keep...	2000-02-26	wk5	87
2 Pac	Baby Don't Cry (Keep...	2000-02-26	wk6	94

https://r4ds.hadley.nz/data-tidy.html

Basic steps

Import → **Tidy /transform** → Visualize

id	bp1	bp2
A	100	120
B	140	115
C	120	125

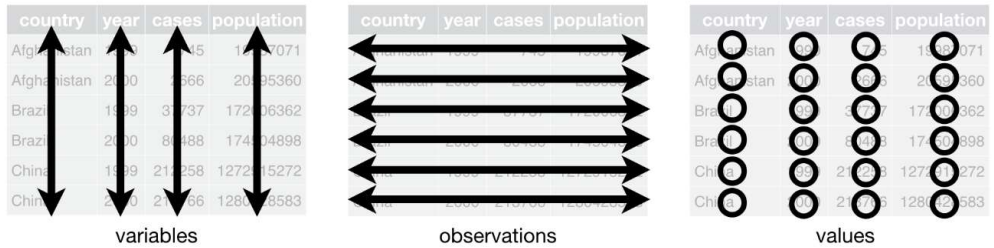
id	measurement	value
A	bp1	100
A	bp2	120
B	bp1	140
B	bp2	115
C	bp1	120
C	bp2	125

id	x_1	x_2	y_1	y_2
A	1	2	3	4
B	5	6	7	8

id	x	y	num
A	1	3	1
A	2	4	2
B	5	7	1
B	6	8	2

id	x_1	y_2
A	1	2
B	3	4
C	5	6

id	name	number	value
A	x	1	1
A	y	2	2
B	x	1	3
B	y	2	4
C	x	1	5
C	y	2	6



https://r4ds.hadley.nz/data-tidy.html

Type of Data

Structured Data

- Numerical: Integers, decimals, percentages, financial metrics
- Categorical: Variables with defined categories (gender, country, product type)
- Ordinal: Categories with specific order (ratings, satisfaction levels)
- Temporal: Dates, times, time series, seasonal data

Unstructured Data

- Text
- Images
- Audio
- Video

school	sex	age	address	famsize	Pstatus	Medu	Fedu	Mjob	Fjob	reason	guardian	traveltime
GP	F	18	U	GT3	A	4	4	at_home	teacher	course	mother	2
GP	F	17	U	GT3	T	1	1	at_home	other	course	father	1
GP	F	15	U	LE3	T	1	1	at_home	other	other	mother	1
GP	F	15	U	GT3	T	4	2	health	services	home	mother	1
GP	F	16	U	GT3	T	3	3	other	other	home	father	1
GP	M	16	U	LE3	T	4	3	services	other	reputation	mother	1
GP	M	16	U	LE3	T	2	2	other	other	home	mother	1
GP	F	17	U	GT3	A	4	4	other	teacher	home	mother	2
GP	M	15	U	LE3	A	3	2	services	other	home	mother	1
GP	M	15	U	GT3	T	3	4	other	other	home	mother	1
GP	F	15	U	GT3	T	4	4	teacher	health	reputation	mother	1

Basic steps

Import → **Tidy /transform** → Visualize

rows
- filter()
- arrange()
- distinct()
- count()

columns
- mutate()
- select()
- rename()
- relocate()

groups
- group_by()
- summarise()

```
#¿Numero de vuelos en cada aeropuerto por año

data5 %>%
  count(Airport.Code,Time.Year) %>%
  head()
```

```
tabla_mes_dia <- data1 %>%
  mutate(
    year = year(fecha),
    month = month(fecha),
    day = day(fecha)
    ) %>%
  group_by(year,month) %>%
  summarise(
    max_day = max(day),
    .groups = 'drop'
  )

print(tabla_mes_dia,n=Inf) #queremos ver todo
```

https://r4ds.hadley.nz/data-tidy.html

Exercises

Using the nycflights13::flights data set:

1. In a single pipeline for each condition, find all flights that meet the condition:
    - Had an arrival delay of two or more hours
    - Flew to Houston (IAH or HOU)
    - Were operated by United, American, or Delta
    - Departed in summer (July, August, and September)
    - Arrived more than two hours late but didn't leave late
    - Were delayed by at least an hour, but made up over 30 minutes in flight
2. Sort flights to find the flights with the longest departure delays. Find the flights that left earliest in the morning.
3. Sort flights to find the fastest flights. (Hint: Try including a math calculation inside of your function.)
4. Was there a flight on every day of 2013?
5. Which flights traveled the farthest distance? Which traveled the least distance?

Data Science

Introduction to Statistical Inference and R for Data Science

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